Research Proposal

RP · MASTER'S APPLICATION

APPLICATION PROFILE / 学生与申请信息

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RESEARCH PROPOSAL

1. Title

Robust Multimodal Representation Alignment for Medical Imaging in Low-Data Settings

2. Abstract

Multimodal learning offers an important way to combine medical images with clinical text, yet existing methods remain vulnerable to representation mismatch when annotations are limited, modalities are imbalanced, or inputs are noisy. This project proposes a robust alignment framework that combines contrastive learning, prototype-based regularization, and cross-modal attention. The framework will be evaluated on public medical image-text datasets under consistent training budgets. Ablation studies, robustness tests, calibration analysis, and cross-dataset evaluation will be used to determine whether each component provides a stable benefit. The expected outcome is a reproducible method and evaluation protocol for multimodal modeling in low-resource medical settings.

Keywords: multimodal learning; medical imaging; contrastive learning; low-data learning; representation alignment

3. Background and Rationale

Medical decisions often rely on images, reports, and structured records at the same time. Mapping these sources into a shared representation space can improve classification, retrieval, and decision-support tasks. However, medical data are expensive to collect and annotate, and different modalities describe the same case at different levels of detail. During my work on a medical image analysis project, I observed that conventional alignment methods became unstable when the number of labeled examples was reduced. This observation motivates a focused investigation of robust representation learning when data are limited.

4. Research Questions

1. Which factors contribute most to unstable image-text alignment in low-data settings?
2. Can prototype-based regularization and cross-modal attention improve contrastive learning under class imbalance?
3. Does the proposed framework remain transferable and interpretable across datasets and downstream tasks?

5. Related Work

Existing research can be grouped into joint embedding methods, contrastive objectives, pretrained multimodal models, and cross-modal attention mechanisms. Contrastive learning can establish a shared semantic space from paired samples, but it is sensitive to negative-sample quality and batch composition. Prototype-based methods can reduce within-class variation, although they may amplify the effect of incorrect labels. Attention mechanisms can identify local relationships between modalities, but their computational cost and stability require careful evaluation. The final version of this proposal should review eight to twelve recent core studies and state clearly how the proposed framework differs from prior work.

6. Methodology

4. Data preparation: select public paired medical image-text datasets; conduct quality checks; define training, validation, and test partitions; and construct controlled low-data subsets.
5. Baseline reproduction: reproduce standard contrastive and attention-based models under a unified preprocessing pipeline, training budget, and evaluation protocol.
6. Method development: introduce class-prototype regularization and cross-modal attention into the contrastive framework, then isolate the contribution of each component.
7. Evaluation: report accuracy, F1, retrieval recall, and calibration error; conduct ablation, robustness, cross-dataset, and qualitative error analyses.

7. Expected Outcomes

- A measurable improvement of approximately two to three percentage points over strong baselines under a consistent experimental setting.
- Reproducible code, model configurations, experiment logs, and an error-analysis report.
- One or two research papers or technical reports that describe both the contribution and the limitations of the proposed method.

8. Timeline

Phase 1 — Months 1–3

Work: Review the literature, define the research scope, and prepare the datasets.

Deliverable: A structured literature review and data specification.

Phase 2 — Months 4–6

Work: Reproduce baselines and establish the evaluation protocol and experiment environment.

Deliverable: A baseline comparison report.

Phase 3 — Months 7–12

Work: Develop the method and conduct ablation and robustness experiments.

Deliverable: Core experimental results and an interim analysis.

Phase 4 — Months 13–18

Work: Complete cross-dataset validation, write the paper, and package reproducible materials.

Deliverable: A final manuscript and reproducible code release.

9. References

[1] [Insert recent core references on multimodal representation learning.]

[2] [Insert references on medical image-text alignment and low-data learning.]

[3] [Format all references according to the target program's required citation style.]